

Spring 2026. Physics 510.

Quantum Theory II.

Test # 3.

Due to April 29th, 2026.

Send me a PDF file with your detailed answer by April 29, 2026. Good luck!

YM theory is a direct and straightforward generalization of the scalar electrodynamics we spoke about in the classroom. The latter is the theory of a scalar doublet, i.e. complex scalar field $\phi = \phi_1 + i\phi_2$, with a localized symmetry group $U(1)$ and a singlet vector field A_μ .

$U(1)$ is the simplest continuous group and, therefore, it sounds absolutely logical to consider a scalar complex multiplet $\phi^a, a = 1, 2, 3 \dots M$, where M is the multiplet dimension and its symmetry group G of the dimension N . It is important to notice that generally speaking $M \neq N$ because M is the dimension of the group G representation the ϕ^a multiplet is transformed on. Yang and Mills did it for the $SU(2)$ case. See the original paper as a pattern for your further calculations.

The invariance of $\phi^a \phi^a = \vec{\phi}^+ \cdot \vec{\phi}$ requires that the scalar field is transformed the following way $\vec{\phi} \rightarrow \hat{U} \vec{\phi}$, (or $\phi^a \rightarrow U^a_b \phi^b$) where $\hat{U} = e^{-i\hat{T}_n \omega_n}$, $\hat{T}_n$ are generators of the group representation, ω_n are parameters of the gauge transformation, and $n = 1, 2, 3 \dots N$.

Consider the Lagrangian $\mathcal{L} = \frac{1}{2} \partial_\mu \vec{\phi}^+ \cdot \partial^\mu \vec{\phi} - \frac{1}{2} m^2 \vec{\phi}^+ \cdot \vec{\phi}$. Now assume that the infinitesimal gauge transformation parameters depend on the coordinates $\delta\omega_n(x)$.

Introduce the gauge-invariant derivative $\mathcal{D}_\mu \vec{\phi} = (\partial_\mu + ig\hat{T}_n A^n_\mu) \vec{\phi}$ instead of the usual one ∂_μ , where A^n_μ is a vector multiplet (Attention, the dimensions of the scalar and vector multiplets are not necessary equal. Why?) and transforms as following: $\delta A^n_\mu = C^{nmk} \delta\omega^m A^k_\mu + \frac{1}{g} \partial_\mu \delta\omega^n$.

Show that the model is locally gauge invariant. You will need to recall connection between a commutator of generators and group structure constants.

Calculate the vector field strength tensor using the following formula for the commutator of the gauge-invariant derivatives applied on the scalar field $[\mathcal{D}_\mu, \mathcal{D}_\nu] \vec{\phi} = ig\hat{T}_n F^n_{\mu\nu} \vec{\phi}$.

Derive the law of $F^n_{\mu\nu}$ transformations under the infinitesimal local gauge transformation. It is not invariant.

Prove that the pure YM field Lagrangian $\mathcal{L}_{YM} = -\frac{1}{4} F^n_{\mu\nu} F^{n\mu\nu}$ is indeed invariant. Notice that the free YM field Lagrange equations are non-linear!

Now look at the original YM paper. It discusses a particular, the most elementary case of a non-Abelian symmetry group, that is $SU(2)$. Recall the structure constants of this group and get the formula (9) of the paper. It is quite common to use $\mathcal{A}_\mu = \hat{T}_n A^n_\mu$ instead of A^n_μ .